Concealed Communication: Investigating Techniques and Algorithms Across Image, Video, and Text Steganography

Esha Sachin Potdar (21BCE1136)
Vellore Institute of Technology (VIT)
Chennai, India

Runi Ghosh (21BCE5217)
Vellore Institute of Technology (VIT)
Chennai, India

Nayoneeka Paul (21BCE5079)
Vellore Institute of Technology (VIT)
Chennai, India

***Abstract*-- As our world becomes increasingly digitized, digital images' security for transmitting critical data is paramount across various sectors. However, ensuring the confidentiality and integrity of these images on the internet poses significant challenges. Steganography and cryptography have emerged to be very instrumental techniques in protecting sensitive information in digital media. More specifically, steganography makes use of a subtle technique that embeds the information in what one might consider innocuous digital files: image, audio, and video files, and even text documents. Commonly employed steganographic methods such as LSB modification, Discrete Wavelet Transform (DWT), Discrete Cosine Transform (DCT), Integer Least Significant Bit (ILSB), and Singular Value Decomposition (SVD) provide diverse avenues for embedding and extracting hidden data. This paper elaborates on the steganographic techniques that have come up and their importance and applicability in current information security. The research into different types of steganography and commonly used techniques of steganography is given equal emphasis with the purpose of drilling deep into the effective methodologies for image and data security in a digitized world.**

Keywords*-- *Steganography, Cryptography, Digital Image Security, LSB Modification, Discrete Wavelet Transform, Discrete Cosine Transform, Singular Value Decomposition, Data Integrity.

I. OBJECTIVES

1. **Evaluate Steganographic Techniques:** Compare and contrast various steganographic methods, including LSB modification, Discrete Wavelet Transform (DWT), Discrete Cosine Transform (DCT), Integer Least Significant Bit (ILSB), and Singular Value Decomposition (SVD), to assess their strengths and weaknesses in terms of data embedding capacity, imperceptibility, and robustness against attacks.
2. **Performance Analysis:** The performance of steganographic techniques will be judged based on quantitative measurements that provide values for the embedding capacity, signal-to-noise ratio, peak signal-to-noise ratio, and mean square error to seek effectiveness in data hiding with the least possible deterioration of the quality of the images.
3. **Security Evaluation:** Explain the security-related implications of each steganographic technique by checking its vulnerability against common attacks through statistical analysis, visual inspection, and attacks in general with steganalysis techniques. Calculate the efficacy of each technique against detection to retain the secrecy of the hidden data.
4. **Visualization and Interpretation:** The results should be displayed with graphs, tables, and other graphical aids to assist the comparative analysis of steganographic techniques. Give insights and interpretation based on results that exhibit performance differences and trade-offs among different methods.
5. **Recommendations:** From the comparative study, recommend how one can choose or select the most appropriate steganographic technique(s) for a particular application, taking into consideration factors such as security level, embedding capacity, computational complexity, and robustness in the face of various attacks.

By delineating these objectives, your comparative analysis aims to provide a comprehensive understanding of the effectiveness and suitability of different steganographic techniques for image and data security purposes.

II. INTRODUCTION

Throughout history, the need to conceal information from prying eyes has driven the development of various techniques. From ancient Greece's attempts to transmit covert messages through hostile territory to modern digital communication, the art of hiding data has evolved. In today's interconnected world, where information security is paramount, techniques

such as cryptography and steganography play vital roles in safeguarding sensitive information. While cryptography focuses on encrypting and securing data during transmission, steganography takes a different approach, aiming to hide the existence of information itself. Stemming from the Greek words "stegos" meaning "cover" and "graphy" meaning "writing," steganography involves concealing data within seemingly innocuous mediums such as digital images. In this introduction, we delve into the concept of steganography, the art and science of secret communication, exploring its history, techniques, and applications in modern communication.

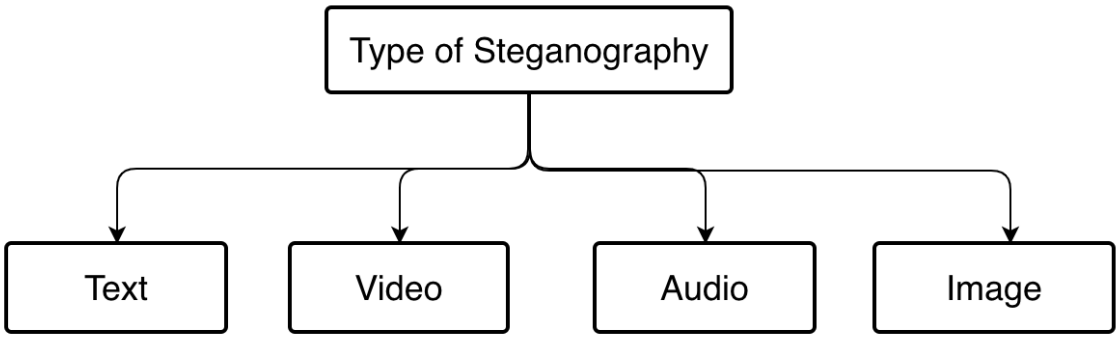


Figure 1: The different types of Steganography

2.1 Types of Steganography :

2.1.1 Text Steganography:

Text steganography (Figure 1) involves hiding secret messages within text or textual data. This method was historically popular for concealing messages within longer text passages, such as in newspapers or books. However, with the advent of the internet and digital documents, its significance has decreased due to the limited amount of extra information that can be concealed within text files.

2.1.2 Image Steganography:

Image steganography (Figure 1) is one of the most straightforward methods of concealing information. It involves embedding a message within a digital image using mathematical algorithms and encryption keys. The resulting stego image appears identical to the original image to the untrained eye. Only authorized individuals with the appropriate decryption key can extract and view the hidden message.

2.1.3. Video Steganography:

Video steganography (Figure 1) entails hiding information within video files. Similar to image steganography, the message is embedded within the video using encryption techniques, making it imperceptible to unauthorized viewers. Video-based steganography techniques utilize methods similar to those used for images and are categorized into spatial and frequency domain methods. Evaluating the performance of video steganography involves considering both its advantages and drawbacks, such as its ability to conceal information without significantly impacting visual quality.

III. SUMMARY OF RELATED WORK

Image steganography involves embedding secret information within digital images for covert communication. This overview synthesizes multiple research papers to provide a comprehensive understanding of its techniques, challenges, and future directions.

Early image steganography techniques primarily focused on spatial and frequency domains. [1, 17] compared Least Significant Bit (LSB), Discrete Cosine Transform (DCT), and Discrete Wavelet Transform (DWT) methods, evaluating them using Peak Signal-to-Noise Ratio (PSNR) and correlation coefficient. While LSB offered simplicity, DCT and DWT demonstrated enhanced security and imperceptibility through frequency domain transformations.

To improve embedding capacity, security, and imperceptibility, researchers explored various techniques. [5] proposed combining steganography with visual cryptography for enhanced security. [6] introduced SCRIPT, a framework for embedding messages in text-to-image generated images, emphasizing robustness and synchronization. [9] presented HIGAN, a deep learning-based model for embedding color images with high visual quality and security.

A novel approach, coverless image steganography, was introduced in [10]. By generating a visually similar image to the secret image, this method eliminates the need for embedding, enhancing security.

Despite advancements, challenges persist in achieving the optimal balance between embedding capacity, security, and imperceptibility. [8, 11, 22] emphasized the importance of comprehensive evaluation metrics, including robustness, security, imperceptibility, and embedding capacity. [19, 20] focused on steganalysis techniques to detect hidden information, highlighting the ongoing arms race between steganography and steganalysis.

To address the limitations of existing techniques, researchers are exploring innovative approaches. [21] proposed using Huffman coding, LSB steganography, Multi-Level Encryption, and the HSI color model for enhanced security and reduced processing time. [27] introduced a biometric-based steganography algorithm, leveraging human hand features for improved security and imperceptibility. The integration of machine learning and deep learning is becoming increasingly prevalent in steganography and steganalysis. [9, 19] demonstrated the potential of these techniques in enhancing embedding and detection capabilities.

Future research should focus on developing robust and adaptive steganography techniques capable of handling diverse image formats and embedding capacities. Exploring hybrid approaches that combine multiple techniques, such as deep learning and traditional methods, holds promise for achieving optimal performance. Additionally, addressing the challenges posed by evolving steganalysis techniques is crucial for maintaining the effectiveness of steganography.

Video steganography is a method of embedding secret information within video files, leveraging their inherent redundancy and widespread use. This technique enhances data security by concealing sensitive data from unauthorized access.

[2, 7, 12, 24] Research in this area explores various methods for embedding secret data in videos, focusing on imperceptibility, hiding capacity, and robustness. These studies delve into both compressed and raw video formats, examining different embedding stages within the compression process, such as intra-frame prediction, inter-frame prediction, motion vectors, transformed and quantized coefficients, and entropy coding. 1 Additionally, spatial and transform domains are explored for raw video steganography.

[2] One particular study proposes a deep learning-based approach for pixel prediction and secret image embedding in H.264 videos. By utilizing an optimized DCNN and the Chronological Gazile optimization algorithm, the method achieves high imperceptibility and robustness, as measured by Structural Similarity Index (SSIM), Normalized Correlation, Peak Signal to Noise Ratio (PSNR), Bit Error Rate (BER), and Mean Squared Error (MSE), with values of 0.979, 0.974, 49.624, 4.655, and 0.790, respectively.

[7, 12, 24] While existing techniques demonstrate promising results, challenges remain in achieving optimal performance across different video formats and data types. [24] suggests that text embedding in videos is generally more secure than image embedding. However, a universal solution capable of handling various video formats (AVI, MOV, MPEG) and data types (text, image, audio, video) is still sought after.

Future research should focus on enhancing compression, decompression, encryption, decryption, and random data embedding processes to improve efficiency, flexibility, and security. By addressing these aspects, video steganography can become a more robust and reliable method for protecting sensitive information.

Text steganography is a method of concealing secret information within plain text documents. [1] It has gained increasing importance due to the growth of internet and mobile communications. This overview combines several research papers to provide a comprehensive understanding of text steganography techniques and their advancements.

Early text steganography techniques primarily relied on exploiting existing text elements like spaces, punctuation, or specific character combinations. For instance, [13] introduced a method using Unicode spaces and zero-width characters, enhancing capacity through compression and bolstering security with SHA-1. This approach demonstrated improved performance compared to previous techniques.

To further enhance security and capacity, researchers delved into language-specific characteristics. [14] proposed the ZKS algorithm for Arabic text, embedding information within Kashida letters and zero-width characters. Similarly, [15,

16] explored English and Arabic text, respectively, utilizing specific character combinations for data hiding. These methods often prioritized maintaining text readability and integrity.

Addressing the limitations of traditional methods, researchers introduced novel approaches. [23] proposed a technique combining POS tagging, randomization, and Format Preserving Encryption (FPE) to enhance embedding capacity and imperceptibility. By utilizing Unicode characters at random positions, this method significantly improved performance compared to traditional techniques.

A groundbreaking development in text steganography involves the integration of neural networks. [25] introduced NMT-Steg, a linguistic steganography methodology employing an attention-based neural machine translation model. This approach demonstrated high embedding rates, preserved text quality, and offered resistance to steganalysis.

While character-based embedding remains a cornerstone, researchers explored alternative approaches. [29] introduced a novel method utilizing text formatting, particularly text color, as a carrier for hidden information. This approach showcased the potential for increased storage capacity and improved security.

To safeguard sensitive information, cryptographic measures are essential. [3] emphasized the importance of threat modeling and cryptographic protocols in designing secure steganography systems. Additionally, efficiency is paramount. [30] proposed optimization techniques to maximize embedding capacity while minimizing file size increases.

Future research aims to further enhance embedding capacity, security, and imperceptibility. This includes developing more robust steganalysis techniques, exploring new embedding carriers, and adapting to evolving text formats and languages. Additionally, integrating steganography with other security mechanisms, such as cryptography and biometrics, can create more comprehensive protection for sensitive information.

In conclusion, steganography, encompassing techniques for concealing information within various media, has evolved significantly to meet the growing demand for secure communication. While advancements in image, text, and other forms of steganography have shown promise, challenges such as balancing capacity, security, and imperceptibility persist. The ongoing interplay between steganography and steganalysis drives continuous innovation in the field. Future research should focus on developing adaptive and robust techniques, exploring novel embedding carriers, and leveraging emerging technologies like artificial intelligence to enhance the effectiveness of steganography in protecting sensitive information.

IV. TABULAR ANALYSIS

Table 1: Image Steganography

Ref	Type of Steganography	Algorithm Name	Pros	Cons	Application	Time and Space Complexity
[1]	**Digital image watermarking:** techniques discussed, such as DWT, DCT, DFT, and SVD, can be applied to other types of digital content like text or video.	Discrete Wavelet Transform (DWT) Discrete Cosine Transform (DCT) Discrete Fourier Transform (DFT) Singular Value Decomposition (SVD)	**Imperceptibility:** Invisible embedding for its detection to be avoided. **Robustness:** it exhibit resistance to attacks like noise addition, compression, and signal processing operations. **Versatility:** Domains used are text, image, and video. **Efficiency:** The combination of DWT and SVD shows enhanced performance with less computational time.	**Complexity**: The techniques are a bit complex combined. **Performance Degradation:** Some methods may cause a slight degradation in the quality of the watermarked content. **Vulnerability:** While robust, these methods are not invulnerable to sophisticated attacks, particularly when the attacker knows the watermarking technique.	**Copyright Protection:** Its used for protecting intellectual property rights in images, audio, and video. **Authentication:** verify the authenticity such as the source of an image or video. **Tamper Detection:** detect and identify any unauthorized modifications. **Fingerprinting:** is used to uniquely mark for identification purposes.	**Discrete Wavelet Transform (DWT):** Time Complexity: O(n) **Discrete Wavelet Transform (DWT):** Space Complexity: O(n)

[2]	The algorithm is specifically designed for **image compression artifacts reduction, particularly in J**PEG images.	Convolutional Neural Network (CNN) based on Discrete Cosine Transform (DCT) domain	**Artifact Reduction:** It reduce compression artifacts, like blocking and ringing, in JPEG images. **Receptive Field Expansion:** It expands the receptive field and eliminate discontinuities between 8x8 blocks by overlapping patch extraction. **Prediction Accuracy:** reduces the loss of JPEG images by more precisely predicting genuine DCT coefficients. **Performance:** reaches cutting-edge performance in terms of both subjective and objective visual quality.	**Complexity:** Implementing and training a CNN based on the DCT domain may be complex and computationally intensive. **Training Data:** The effectiveness of the algorithm may depend on the quality and quantity of training data available. **Generalization:** The algorithm's performance on images with different characteristics or compression levels may vary.	**Image Compression:** Improving the quality of JPEG compressed images, particularly in applications where high-quality images are required, such as medical imaging or satellite imaging. **Digital Photography:** Enhancing the visual quality of compressed images in digital cameras or smartphones. **Video Streaming:** Improving the quality of compressed video streams to enhance viewing experience.	**Forward Pass:** O(n **Backward Pass (Gradient Calculation):** O(n). **Training:** O(m * n * e), where m is the number of training samples, n is the number of operation per sample, and e is the number of epochs. **Inference (Prediction):** O(n).
[3]	The algorithm is designed for hiding secret messages in images, **so it primarily applies to image data.**	Steganography combined with Visual Cryptography	**Enhanced Security:** . Improved security by steganography and visual cryptography provides double layers of protection. **Reliability:** in that the message can always be recovered with visual cryptography, even if some shares are lost. **Efficiency:** It is efficient since least significant bit hiding interferes as little as possible with the picture quality. **Practica**lity: in the sense that its MATLAB implementation makes it accessible and easy to use.	**Complexity:** Implementing the combined steganography and visual cryptography algorithm can be complex and may require a deep understanding of both techniques. **Detection:** While the method aims to make the detection of the hidden message difficult, it may still be susceptible to advanced detection techniques.	**Secure Communication:** The method can be used to securely transmit sensitive information over insecure channels, such as email or messaging apps. **Digital Watermarking:** For copyright protection and authentication, digital watermarks can be embedded into photos using this technique.**Data Privacy:** The method can be applied to protect personal and confidential data stored in images.	**Embedding:** O(n), where n is the number of pixels in the image. **Visual Cryptography:** O(n where n is the number of pixels in the image. This includes the overhea of generating noise-like structures for each share. **Embedding:** O(1), a it involves modifyin the least significant bits of the original image. **Visual Cryptography:** O(n as it requires storing noise-like structures for each share.
[4]	**Primarily image**, but also involves text (prompts) and potentially a shared random seed.	SCRIPT	High security (20% improvement) Robustness against unknown image alterations High message extraction success rate (98%)	Requires sharing the prompt alongside the image, potentially raising suspicion. Relies on a mutually agreed-upon random seed, introducing a potential point of failure.	Covertly transmitting information within seemingly innocuous AI-generated social media posts, potentially for espionage or other malicious purposes.	Proportional to the image size (O(n), where n is the number of pixels) in the worst-case scenario. the space complexity is likely constant (O(1)) in terms of th message size.

[5]	The paper focuses on **image steganography,** although it might mention other file types and briefly discuss other security techniques.		Emphasizes the importance of data security in the digital era. Introduces steganography as a potential tool for secure communication. Provides an overview of various image steganography techniques. Discusses potential applications of steganography in various fields.	Doesn't offer in-depth analysis of individual steganographic techniques. Doesn't discuss the limitations or ethical concerns associated with steganography.	The paper mentions applications in: **Defense and intelligence:** Secure communication in sensitive operations. **Medical field**: Protecting patient privacy by embedding medical data in images. **Online transactions:** Securing sensitive financial information. **Copyright protection:** Embedding watermarks in digital content.	
[6]	HIGAN is primarily designed for **image steganography.**	HIGAN (High-quality Image steganography with Generative Adversarial Network)	**Improved Image Quality:** Uses residual blocks to minimize distortion and maintain high visual quality of stego-images. **Enhanced Security:** Employs adversarial training to resist deep learning-based steganalysis models. **Increased Capacity:** Potentially allows for hiding larger amounts of data compared to previous models.	**Complexity:** Implementing and training deep learning models like HIGAN can be computationally expensive and resource-intensive. **Potential Ethical Concerns:** Steganography, while having legitimate uses, can also be misused for malicious purposes like covert communication in illegal activities.	HIGAN has the potential to be used in various applications requiring secure data transmission, such as: **Military or law enforcement communication:** Securely sharing sensitive information without raising suspicion. **Medical data protection:** Embedding confidential medical data in images for secure storage or transmission. **Copyright protection:** Embedding robust watermarks in digital content to prevent unauthorized use.	Training HIGAN, which involves a deep learning model with residual blocks and adversarial training, is likely highly complex and time-consuming. Training deep learning models ofte requires iterating through large dataset multiple times, making the process computationally expensive. Training HIGAN als likely requires significant storage space for several reasons: Storing the model parameters itself can consume a significan amount of memory. Training often involves keeping a large dataset of images for the learning process.

[7]	The text primarily focuses on **image steganograph**y, although it mentions "meaning-normal and independent image" which might involve some level of text processing within the generative model.	a unique generative model-based coverless image steganographic technique.	**High Security:** Steganalysis techniques have a harder time detecting the secret image as the transmitted image doesn't directly reveal anything about it. **High Capacity:** Potentially allows for hiding more information compared to traditional methods. **No Cover Image Needed:** Eliminates the need for a separate cover image, potentially simplifying the process.	**Reliance on Generative Model:** Requires access to a trained generative model, which may not be readily available. **Computational Complexity:** Training and using generative models can be computationally expensive. **Potential Security Concerns:** The security of the scheme depends heavily on the robustness of the generative model and its database.	Applicable in: **Covert communication:** Discreetly transmitting sensitive information without raising suspicion. **Secure data transfer in restricted environments:** Sharing confidential data in scenarios where using traditional cover images might be impractical.	
[8]	The paper focuses on **image steganography,** although it might mention other file types and briefly discuss other security techniques.	LSB, transform domain (DCT, DWT), spread spectrum, statistical, distortion, file embedding, palette embedding, and adaptive steganography	1. **LSB techniques** offer high payload capacity. 2. **Transform domain techniques** provide better robustness against statistical attacks. 3. **Spread spectrum techniques** are robust against detection. 4. **Statistical techniques** can hide data effectively depending on the cover image. 5. **Distortion techniques** do not affect the statistical properties of the image. 6. **File and palette embedding** offer high payload capacity.	1. LSB techniques are vulnerable to statistical attacks and image manipulation. 2. Transform domain techniques may have lower payload capacity. 3. Spread spectrum techniques may be compromised with digital processing. 4. Statistical techniques are vulnerable to certain attacks and depend on the cover image. 5. Distortion techniques may limit payload capacity and require secure transmission of the cover image.	Spread spectrum encoding is extensively used in **military communications** due to its robustness against detection.	**LSB Insertion:** O(n), where n is the size of the cover media, for both time and space. **Transform Domain Techniques**: Time complexity O(n log n) and space complexity is O(n). **Statistical Steganography:** High time and Space complexity is O(n). **File Embedding and Palette Embedding:** O(n) for time and space. **Adaptive Steganography:** Time complexity O(n log n) and Space complexity is O(n). **Spread Spectrum:** Space complexity is O(n). **Distortion Steganography:** Space complexity is O(n) for storing the stego-object.

[9]	**Image Steganography**, where the hidden data is embedded within digital images.	The paper compares three main image steganography algorithms: **LSB Substitution:** The message bits are inserted in lieu of the least important pixels in the cover image. **Discrete Cosine Transform (DCT):** hides data by modifying the coefficients obtained from applying the DCT to the image. Discrete Wavelet **Transform (DWT):** it transforms the image to the frequency domain using wavelets. Here, secret data is embedded in the high-frequency coefficients.	**Least Significant Bit (LSB) Substitution:** Simplicity Low Computational Imperceptible Changes **Discrete Cosine Transform (DCT) Transform:** Higher Capacity Improved Robustness Compatible with JPEG	**Least Significant Bit (LSB) Substitution:** Low Capacity Low Robustness Vulnerable to Lossy Compression **Discrete Cosine Transform (DCT) Transform:** Increased Complexity Potential Image Quality Loss Vulnerable to Specific Attacks **Discrete Wavelet Transform (DWT) Transform:** Highest Complexity Potential Image Quality Degradation	No Real-Life Applications mentioned in paper	
[10]	The paper focuses on **Image Steganography**, which hides messages within images.	**LSB (Least Significant Bit):** In order to implant the secret data, the least important portions of the image pixels must be changed. **DWT (Discrete Wavelet Transform):** DWT breaks down the image into different frequency bands, potentially for hiding data in specific bands. DCT (Discrete Cosine Transform): DCT transforms the image into the frequency domain, where data can be hidden. Other algorithms used in other research (not necessarily this paper) are also mentioned, like SVD, K-Means Clustering, and Hashing.	High Secrecy Improved Security for Encrypted Messages Increased Transmission Capacity Legal Restrictions on Encryption	Reduced image quality: Hiding data can alter the original image slightly, potentially reducing its quality. Limited hiding capacity: The amount of data you can hide in an image using steganography is limited. Detectability: Sophisticated techniques can potentially detect the presence of hidden data.	No Real-Life Applications mentioned	

[11]	**Image steganography**: steganographic techniques such as WOW, S-UNIWARD, HUGO, and MiPOD.	Curvelet Transform: This technique used to extract features from images, particularly suited for analyzing curvilinear structures. **Support Vector Machine (SVM):** This algorithm used for classification tasks, like used to classify images as containing a hidden message (stego-image) or not (cover-image).	The paper claims the proposed method can effectively extract features from both low-payload and high-payload stego-images using Curvelet transforms. SVM classifier known for good accuracy. The method might be suitable for cloud environments due to its potential for accurate stegano detection.		No Real-Life Applications mentioned in paper.	
[12]	The paper focuses on **image steganography**, where hidden messages are embedded within image files.	Steganography Algorithms (Embedding) F5 (widely used algorithm) Outguess Steghide Steganalysis Algorithms Signature-based techniques Statistical techniques AI-based techniques (possibly machine learning like SVM) Aletheia (a tool using various techniques) StegExpose (another tool)	**AI-based techniques:** Potentially higher accuracy when the steganography algorithm is unknown (blind steganalysis). **Statistical/Signature-based techniques:** High accuracy when the specific steganography algorithm used is known. **Aletheia:** More reliable tool compared to StegExpose in the context of the research. **Steghide:** Among the examined tools, Steghide maintains image quality (PSNR).	**AI-based techniques:** Not necessarily better than traditional techniques and may not guarantee high detection in all cases. **StegExpose:** Low accuracy (50% false negative rate) in the tested scenario. **PSNR, MSE, Color, Image Size:** Not reliable indicators of steganography in JPEG images (for the tested tools).	The paper mentions Digital Forensic Investigation (DFI) as a potential application domain for steganalysis techniques. DFIs aim to uncover digital evidence from devices/storage, and steganalysis can help detect hidden messages used for malicious purposes.	

[13]	**Image steganography**	Multi-Level Encryption (MLE) Huffman encoding and decoding Magic Matrix operations Least Significant Bit (LSB) substitution	**Enhanced Security:** Multi-level encryption (MLE) and Huffman encoding contribute to increased security by making the extraction of the actual message more challenging for potential attackers. **Reduced Computational Overhead:** The goal of using the HSI image's I-plane instead of the RGB model is to save computational overhead without sacrificing safety. **Improved Robustness:** The introduction of Magic Matrix makes steganalysis process more complex, improving robustness of steganographic method. Better Quality Stego Images	**Complexity:** Some of the algorithms, such as Multi-Level Encryption (MLE) and Magic Matrix operations, might introduce complexity into the implementation and understanding of the steganographic method **Resource Intensiveness:** Certain algorithms may require significant computational resources, potentially impacting the efficiency of the steganographic process.	Doesn't explicitly mention any real-life applications	**Huffman coding:** Time and space complexity: O(n *log n) and O(n) where n is the size of the data being compressed / decompressed. **HSI color model conversion:** Time and space complexity: O(n), where n is the number of pixels in the image. **Multi-Level Magic matrix:** O(n) as the magic matrix operation requires temporary storage proportional to the data size. **LSB substitution:** Time and space complexity: O(n), where n is the number of pixels in the image and modified image requires the same amount of memory a the original image.
[14]	**Digital Image steganography**	The paper doesn't explicitly mention specific algorithms used for image steganography. Instead, it provides a comprehensive review of existing types of image steganography techniques, including recent contributions.	**Imperceptibility:** This refers to the ability of a steganographic system to hide secret data within a cover media in such a way that the alterations made to the cover media are imperceptible to human observers. **Security:** A steganographic system should provide security to the hidden data, ensuring that it remains confidential and cannot be easily extracted or deciphered by unauthorized parties. **Capacity of hiding information:** This refers to the amount of secret information that can be concealed within the cover media without significantly degrading the cover media's quality or raising suspicion.	**Detection and Analysis:** imperceptibility is a property of steganographic systems, its also a challenge for detecting the presence of hidden information. **Trade-offs between Properties:** The paper mentions the challenge of maintaining a fair trade-off between robustness, security, imperceptibility, and embedding capacity. Enhancing one aspect of a steganographic system may come at the expense of others. **Incomplete Fulfillments of Requirements:** While significant research in this area has been going on, high security, robustness, and large payload capacity all together are still difficult to obtain.	The paper mentions real-life applications - **Medical Field:** it is used to hide crucial private information within medical data or DNA sequences to prevent unauthorized access or leakage of sensitive details. The paper suggests that reversible steganographic systems are more common in this context, likely due to the need to maintain data integrity while concealing sensitive information. **Military:** The paper implies that steganography can be used to conceal sensitive information during communication to prevent interception or detection by adversaries.	The paper doesn't explicitly mention specific time and space complexity analyses for steganography algorithms or techniques. Instead, focuses on discussin the concepts, classifications, properties, and real-life applications of steganography, along with performance evaluation methods. The paper doesn't explicitly mention specific time and space complexity analyses for steganography algorithms or techniques. Instead, focuses on discussin the concepts, classifications, properties, and real-life applications of steganography, along with performance evaluation methods.

[15]	Hiding information within **images**	Technique of Least Significant Bit (LSB) Substitution Spread Spectrum Discrete Wavelet Transform (DWT) Discrete Cosine Transform (DCT) Hash-Least Significant Bits (Hash-LSB)	**LSB Substitution Technique:** High capacity for data hiding Simple implementation **Discrete Cosine Transform (DCT):** Bandwidth reduction through compression. High security and PSNR. **Discrete Wavelet Transform (DWT):** High data hiding capacity. High security and robustness. **Spread Spectrum:** Increased signal bandwidth improves data reception probability. Medium robustness and PSNR. **Hash-Least Significant Bits (Hash-LSB):** Very good MSE and PSNR.	**LSB Substitution Technique**: Low robustness, vulnerable to attacks like low-pass filtering and compression. **Discrete Cosine Transform (DCT):** Limited capacity for hiding large amounts of data. This may result in lower PSNR. **Discrete Wavelet Transform (DWT):** Higher computational cost and Lower PSNR compared to other methods. **Spread Spectrum:** Medium robustness. Potential issues with improving embedded signal estimation.	The paper mentions several real-life applications of the steganography methods described, including: Secure transmission of images over the internet, such as via email or social networking sites. Use in defense and civil applications. Application in space technology for secure transmission of satellite images. Contribution to overall national security efforts.	**LSB substitution**, time complexity is generally low since i involves simple bitwise operations on pixel values of the image. However space complexity may be higher due to the need to store both cover and embedded message. **DCT and DWT** are complex mathematical operations and require higher time complexity, especially for larger images. And may require more memory for intermediate results so higher space complexity. **Spread Spectrum** techniques is spreading the message signal across a wide frequency bandwidth, increasin time complexity and also increase space complexity.

[16]	**Spatial domain steganography** employing the 3-3-2 LSB to hide data within color channels of **digital images**	**Data Concealment Algorithm:** Designed to hide confidential data within digital images based on geometrical dimensions, specifically the engineering dimensions of the human hand. **Data Extraction Algorithm:** This algorithm is used to extract the hidden data from the cover image after it has been received.	**High Storage Capacity:** used for hiding confidential data within digital images. **Efficient Hiding:** ensures that the concealed data is not easily detectable. **Unnoticeable Concealment:** it does not noticeably affect the quality of the image, making it difficult for unauthorized individuals to detect the hidden data. **Integration of Concealment and Biometrics:** It integrates concealment features with biometrics features, enhancing security and preventing unauthorized access to confidential data. **Adaptability to Other Biometrics:** While the algorithm primarily utilizes geometrical dimensions of the human hand, it can be adapted to use other types of biometrics for concealment.	The paper does not explicitly mention any disadvantages or drawbacks of the proposed algorithm.	The paper mentions real-life applications- **Biometric-Based Data Security:** It integrates concealment features with biometric dimensions, enhancing data security by leveraging unique human characteristics. **Preventing Unauthorised Access:** By hiding confidential data within digital images based on geometric dimensions, the algorithm aims to prevent unauthorized access to sensitive information. **Multimedia Concealment**: not only to images but also can be applied to audio, video, etc. **Replacement of Biometric Features:** It can replace geometric dimensions with other biometric features, indicating flexibility in its application.	The time complexity of the proposed algorithm depends o the size of the image and the amount of data being concealed In terms of space complexity, it depends on the storage required for the original image, the hidden data, and any additional data structures used durin the concealment process.
[17]	**Image steganography**	Spatial Domain Technique Masking and Filtering Transform Domain Technique Distortion Techniques	There's a minor chance for degradation of the original image. Higher hiding capacity: More robust compared to LSB replacement concerning compression. Hid information in areas of the image less exposed to compression, cropping, and image processing. Can store a significant amount of data. Provides a strong degree of tamper resistance.	Less robust: hidden data can be lost with image manipulation. Can be applied to grayscale images and restricted to 24 bits. Requires knowledge of original cover image during decoding. Limited benefits due to the need for sending the cover image.	Digital watermarking for intellectual property protection. E-commerce transactions for secure verification.	The time complexity of the algorithm is primarliy O(n), wher n is the total number of pixels in the image. The space complexit is O(m+k), where m is the size of the image and k is the size of the hidden message.

Table 2: Video and audio steganography

Ref	Type of Steganography	Algorithm Name	Pros	Cons	Application	Time and Space Complexity
[1]	The method is specifically designed for **video steganography**, particularly in H.264 videos.	Deep Convolutional Neural Network (DCNN) Chronological Gazelle optimization algorithm (CGOA) Haar Wavelet Transform (HWT)	**Security:** The use of deep learning-based pixel prediction and CGOA optimization enhances the security of the steganography method. **Efficiency:** Achieves high performance in terms of imperceptibility and robustness, as indicated by evaluation measures such as SSIM, NC, PSNR, BER, and MSE. **Applicability:** This can be applied to H.264 videos, which are commonly used in various applications.	**Complexity:** Implementing deep learning algorithms and optimization techniques can be complex and require significant computational resources. **Dependency on Training Data:** The performance of the DCNN depends on quality and quantity of the training data.	**Defense Domain:** Its useful in the defense domain, suggesting applications in secure communication sharing among defense agencies. **Media and Entertainment:** secure distribution of video content, such as movies and TV shows. **Surveillance Systems:** Enhancing security in surveillance systems by embedding information securely in video streams.	**Haar Wavelet Transform (HWT):**O(n) **Haar Wavelet Transform (HWT):**O(n)
[2]	The paper focuses on **video steganography**, although it may mention text descriptions or specific algorithms associated with video data.		Provides a comprehensive analysis of various video steganography methods. Categorizes methods based on compression domain usage and data hiding techniques. Discusses performance evaluations of different methods. Suggests potential research directions for improvement.	Doesn't offer a specific solution or tool. Primarily provides an overview rather than an in-depth analysis of individual methods.	This research directly impacts the development of steganographic tools used in various security applications: **Covert communication** in law enforcement, military to secure transmission of confidential data Digital watermarking for copyright protection.	
[3]	We focus in this paper on **digital audio steganography**, which has emerged as a prominent source of data hiding across novel telecommunication technologies such as covered voice-over-IP, audio conferencing, etc.	**Temporal Domain:** Least Significant Bit (LSB) modification. **Transform Domain:** Echo Hiding Spread Spectrum Phase Coding	Temporal Domain: Simpler to implement, offers high hiding capacity. Transform Domain: More robust due to exploiting masking effects in human hearing. (e.g., echo hiding, spread spectrum, phase coding) Coded Domain (In-Encoder): Most robust, hides data within the codebook of codecs for voice calls or streaming.	Temporal Domain: Less robust, susceptible to noise and manipulation. Transform Domain: Requires careful control over hidden data to avoid detection. Coded Domain (In-Encoder): Vulnerable to transcoding in networks and voice enhancement algorithms. Limited hiding capacity for imperceptibility.	**Secret Communication:** Hiding messages in audio for covert communication (e.g., medical record transfer). **Improved Communication:** Embedding data to enhance the intelligibility of phone calls (e.g., wideband to narrowband conversion). **Data Storage:** Hiding data in audio files for storage (e.g., subtitles in movies).	The paper does not consider these aspec in detail since this is survey. The complexity varies depending upon the technique; for example, temporal methods are relatively simpler compared to transform domain methods.

[4]	Hiding data within **digital videos**	Hash-Based Least Significant Bit (HLSB) Technique LSB Substitution Using Different Polynomial Equations (LSB Poly) More Secured LSB Method (MLSB) Hybrid Encryption and Steganography (HES) LSB Matching Revisited (LSBMR) Algorithm	**Increased Scope for Hiding Data:** Videos offer a larger capacity for hiding secret data due to their nature, which includes numerous redundant bits. **Secrecy:** ensures extraction of hidden information from the video is only possible with prior permission and a specific password. **Imperceptibility:** The ability to be completely undetectable ensures that the presence of hidden data within the video is not easily discernible. **High Embedding Capacity:** Techniques such as LSB substitution allow for a significant amount of data (payload) to be embedded into the cover video. High-Security Applications, Accurate and Reliable.	The paper doesn't explicitly list the disadvantages associated with video steganography techniques. However, some potential drawbacks or challenges may include: Detection Risk: While the techniques aim to be undetectable, there is always a risk that advanced detection methods could uncover the hidden data. Loss of Video Quality: In some cases, embedding data into video files could lead to a loss of video quality or introduce artifacts, if not done carefully.	It doesn't explicitly mention real-life applications	**LSB Steganography Techniques:** (O(n), where n is the number of pixels or frames). **Hybrid Encryption and Steganography (HES):** It depends o the encryption algorithm used and the complexity of embedding encrypte data into the video. **Frequency Domain Transform:** Techniques like Haa Wavelet Transform (HWT) may have a time complexity higher than linear du to the computational cost of performing the transform operation. Also it may require additional memory for storing intermediate.

Table 3: Text steganography

Ref	Type of Steganography	Algorithm Name	Pros	Cons	Application	Time and Space Complexity
[1]	The system is likely applied to **text-based authentication systems**.	Cryptographic methods (both asymmetric and symmetric) Steganography Physiological and behavioral biometrics (not specified)	**Security:** Steganography combines cryptographic methods with biometrics to eliminate passwords and reduce breaches. **Usability:** The system aims to maintain usability by providing authentication within a timeframe similar to passwords (within 2 seconds) and without imposing additional hardware costs on users. **Scalability.** **Privacy.**	**Complexity:** Implementing and integrating cryptographic methods, steganography, and biometrics can be complex and require careful design and implementation. **Compatibility:** The system's compatibility with existing authentication systems and infrastructures may pose challenges. **User Acceptance:** The acceptance of the system by users may vary.	**Enterprise Authentication:** enhance security and usability for employee authentication. **Consumer Authentication:** used in consumer-facing applications like online banking to improve security and user experience. **Government Authentication:** The secure access to sensitive information and services is ensured.	

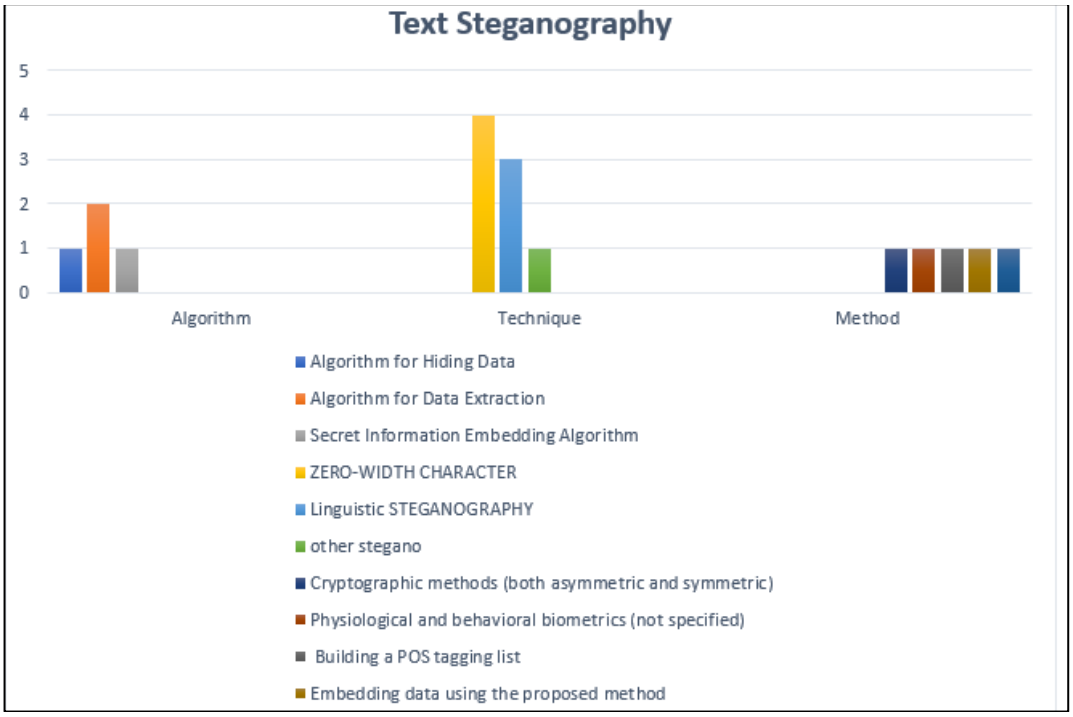
[2]	**Text steganography:** The secret message hidden within a cover text file (Here, MS Word document).	It describes the proposed method that utilizes a combination of techniques: Unicode space characters for embedding secret message bits. Zero-width characters (ZWC) for additional embedding capacity. Permutation and complementation for scrambling the message for security. Compression technique (likely a custom method) to reduce message size before embedding.	**Higher capacity:** Embeds 4 bits per space compared to Unispach's 2 bits/space due to compression. **Smaller stego-text size:** Requires less space in the cover text for the same secret message size compared to existing methods. **Robustness:** Resistant to various attacks like visual, statistical, and format attacks. Notifies the receiver of tampering attempts. **Imperceptibility:** Stego-text appears identical to the original cover text (based on the Jaro-Winkler score).	Increased file size due to hidden data. Detectability by sophisticated steganalysis techniques. Reliance on the security of the embedding method.	The paper doesn't discuss specific real-life applications.	Formal analysis of time and space complexity has not been done in the paper. Compression and specially developed algorithm raise the complexity compared to relatively simpler steganography techniques.
[3]	**Text Steganography :** The algorithm hides data within another text document.	The main algorithm is ZKS (Zero Width and Kashidha Letters). It relies on manipulating Kashida characters and inserting zero-width characters.	Increased hidden data capacity: By using a combination of Kashida with or without zero-width characters, it can hide 2 bits per applicable character compared to 1 bit in prior methods. Improved Steganalysis resistance: Parallel connection fragments the message making it harder to detect. Permutation and randomization further complicate the process for attackers.	The paper doesn't explicitly mention disadvantages, but potential drawbacks could be: Increased file size: Adding Kashida and zero-width characters might slightly inflate the file size. Fragile to some attacks: Techniques like retyping the stego-text (text containing hidden data) could remove the hidden message.	**No Real-Life Applications mentioned**	Time complexity: Fragmentation, permutation, and randomization likely increase processing time compared to simpler algorithms. The complexity might be proportiona to the size of the hidden message. Space complexity: The additional hidde bits and potentially increased file size suggest a slight increase in space complexity compare to the original text file.
[4]	This is **text steganography**. It hides data within a text file.		Perceptual transparency Hiding capacity Robustness	Increased file size Capacity limitations	No Real-Life Applications mentioned	**Time complexity** might be linear (related to the length of the text) fo both embedding and extraction as they likely involve iterating through the text. **Space complexity** is likely also linear. It might require additional space to store temporary information during processing, but the final stego-text file size increase is the main concern.

[5]	**Text Steganography :** This technique hides data within text files.	Algorithm used: Zero Width Characters (ZWC) and space manipulation to hide data.	Preserves file format: ZWCs retain original file format unchanged, making it less susceptible to detection by steganalysis tools that look for format alterations. Unicode/ASCII compatibility: It with various languages as it utilizes Unicode or ASCII characters. Minimal file size impact: hidden data doesn't affect size	The paper doesn't explicitly mention any disadvantages but steganography in general has limitations like: Reduced hiding capacity compared to other media. Potential vulnerability if the steganographic method becomes widely known.	No Real-Life Applications mentioned in paper.	**Mentioned but not quantified:** the time complexity relates to the number of carrier files (files to hide data in) and the number of bits to be embedded. **Not mentioned:** The paper doesn't explicitly discuss the space complexity of the algorithm.
[6]	**Text steganography**	Building a POS tagging list. Embedding data using the proposed method. Extraction process of the proposed method.	**High Capacity:** embedding secret data into cover text files by utilizing part-of-speech tagging and format-preserving encryption techniques. **Improved Security:** ensures data security by encrypting the secret message without modifying size/format. **Unicode Characters:** to hide data bits, allowing for efficient embedding of data while maintaining the steganography system's security and imperceptibility. **Randomization:** The method employs part-of-speech tagging to introduce randomization in selecting positions for embedding secret data, enhancing security against detection. **Maintained Imperceptibility:** Despite high capacity and security enhancements, the method maintains imperceptibility by minimizing the visibility of the cover text.	**Increased File Size:** Due to the use of non-printing Unicode characters to hide data bits, the size of the stego file increases, which may impact storage and transmission efficiency. **Complex Implementation:** Implementing the proposed method may require a level of complexity, particularly in integrating part-of-speech tagging and format-preserving encryption algorithms. **Dependency on Language:** The effectiveness of the technique may depend on the language and the availability of part-of-speech tagging tools for that language. **Limited Imperceptibility Evaluation:** The paper mentions imperceptibility analysis using Jaro–Winkler distance, but it may not provide a comprehensive assessment of the method's perceptual quality.	The paper mentions real-life applications of the proposed text steganography technique. Specifically, it suggests that the method could be applied in communication systems, where it's necessary to encrypt certain protocols, like in in industrial or military situations. Additionally, it highlights the potential use of the proposed technique in improving the security level of stego text files while maintaining capacity and imperceptibility, indicating its relevance in practical scenarios where data security and confidentiality are crucial.	Building **POS tagging list** -Time and Space Complexity: O(n + m), where n is the number of words in the cover text and m is the number of tokens. O(n + k), where n is size of the cover text file and k is number of unique tags. **Embedding data**: Time and space Complexity: O(n + m) and O(n + k), where n size of input text files and m is the number of tokens that are present in the cover text and k is the number of unique POS tags present. **Extraction:** Time and Space Complexity - O(n + m) and O(n + k), where n is the size of the cover text and m is the number of tokens in the cover text. Extraction process of the proposed method Complexity, where n is size of the cover text and k is the number of unique UCS or POS tags present.

[7]	Linguistic steganography [**Text-based**]	Secret Information Embedding Algorithm Secret Information Extraction Algorithm	**Dynamic Embedding:** dynamically embedding secret information during the text generation process, maintains semantical connections with distant words. **Adaptive Embedding Policy:** The policy adjusts the embedding capacity based on the current conditional probability distribution, enhancing adaptability. **Attention-based Adjustment:** degree of dependency between the target sentence and the source sentence, allowing for fine-tuning of the embedding process. **Semantic Relevance:** guarantees coherence; considers source text and previous target words. **Improved Performance:** language model performance metrics (PPL, BLEU score) improve text quality and translation accuracy.	The paper doesn't explicitly mention any disadvantages of the proposed model.	The paper mentions potential real-life applications for the proposed NMT stage model. It suggests technology could be applied in various fields, like improving steganalysis, enhancing information security, or advancing other applications of steganography.	**Training Phase:** Time-consuming because the model learns from a lot of data, influenced by the model's complexity and dataset size. **Inference Phase:** Time varies based on sentence length and model efficiency. **Word Selection and Encoding:** Depends on the number of options the model ha and its decision-making speed. **Space complexity** is determined by the model's size, data volume, and computational steps.
[8]	**Text steganography**	It aims to hide a message within a text file by manipulating the color of text characters, ensuring that the visual integrity of the text is maintained while allowing the detection of the hidden message using specialized software.	The algorithm ensures that the visual integrity of the text is maintained, that is to the human eye, the text appears unchanged, making it difficult for anyone to suspect that it contains hidden information. Using multiple colors for encoding, lets larger amounts of information to be hidden within the text without altering it. The algorithm can hide messages efficiently within text files, with reasonable execution times.	While using color for encoding increases capacity, the paper acknowledges that capacity is directly proportional to the number of colors used. This implies that there could be limitations on the amount of information that can be concealed within the text, especially if the color range is restricted. If the text format does not support the required features, then thr algorithm may not be applicable or may not perform optimally.	The applications are secure communication and information hiding, where sensitive data can be transmitted covertly within seemingly innocuous text documents. Furthermore, the paper mentions the use of the algorithm with both Devanagari and Latin scripts, indicating its potential applicability across different languages and writing systems.	The time complexity of encoding the message into the text file would depend on the size of the input file and the number of bits to be hidden. The space complexit primarily depends upon the size of the input file and the additional space required to store the hidden message within the text.

[9]	**Text steganography**	Algorithm for Hiding Data Algorithm for Data Extraction	The algorithm ensures that the format of the carrier text file remains unchanged after embedding hidden data. There is language independency (use of Unicode or ASCII coding.), enhancing the flexibility and usability of the algorithm across different linguistic contexts. It enhances transparency, helping in concealing the presence of hidden data. It's optimization techniques minimize changes in file size while increasing the hiding capacity of the data, thus enables users to hide a significant amount of data between two letters in a text file.	Traditional steganography techniques were often larger in file sizes, which raised suspicion during transmission. In image steganography, the replacement of certain bits can lead to image distortion, making it easier for steganalysis techniques to detect the presence of hidden data. Text files lack the redundant patterns often found in image, audio, or video files, making them challenging carriers for steganographic data hiding.	Text steganography offers a means of secure communication by concealing messages within seemingly innocuous text. This can be valuable for individuals or organizations requiring secure communication channels, such as government agencies, intelligence services, or businesses handling sensitive information. In scenarios where users need to communicate covertly or under surveillance, text steganography enables the concealment of messages within everyday communication channels.	The time complexity of the hiding algorithm involves iterating through the carrier file and embedding the hidden data, that is linear concerning the size of the carrier file and the length of the hidden data. The time complexity of the extraction algorithm involves iterating through the file and extracting th hidden data, that is linear concerning the size of the file. The space complexit depends on the size of carrier file, length of the hidden data used during the embedding and extraction processes, and is constant or linear concerning the size of the carrier file and the length of the hidden data.

V. GRAPHICAL ANALYSIS



Graph 1: Comparision and analysis of Text Steganography algorithms, techniques and methods.

Note:

Graphical Interpretation :

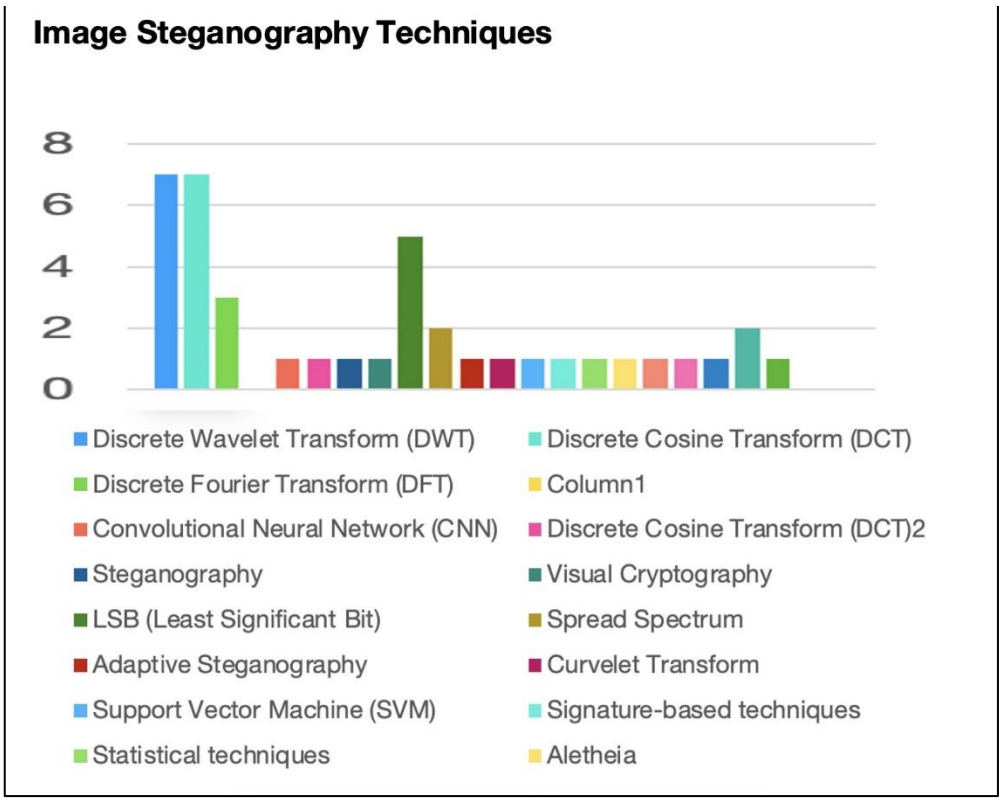
1. Text Steganography (Table 3):

The Zero-Width Character Technique is the most used steganographic technique since it can supply secret communication without any kind of perceivable change in the text (Graph 1). Zero-width characters are invisible to the human eye; therefore, they become extremely useful for hiding messages within texts without arousing suspicion. These characters enjoy very great support from text-processing systems and are therefore portable across all platforms, from text editors and web browsers down to messaging applications. This flexibility makes it possible to hide all types of content, be it binary data or simple text; hence, it is a preferred option for backchannel communications.

The extraction of data in steganography provides the basis upon which reversibility, reliability, and usability of the hidden information are founded. The interest this sparks into the topic has made possible the exploration of effective methods for the hiding and retrieval of hidden messages, hence advancing this field and its applicability.

Though zero-width characters are very popular, some of the other methods which have been mostly used are linguistic steganography. Linguistic steganography embeds secret in the natural language text through what is said and how that is said by using those exact same aspects which create the spectrum from art to noise: word choice, syntax, and punctuation. This technique is unnoticeable and convenient because it does not require any special carriers or encoding techniques.

Different steganographic methods would be appropriate for different contexts and requirements. Actually, each of them had its benefits and limitations depending on the exact application, security needs, and environment conditions



Graph 2: Comparision and analysis of Image Steganography's different techniques.

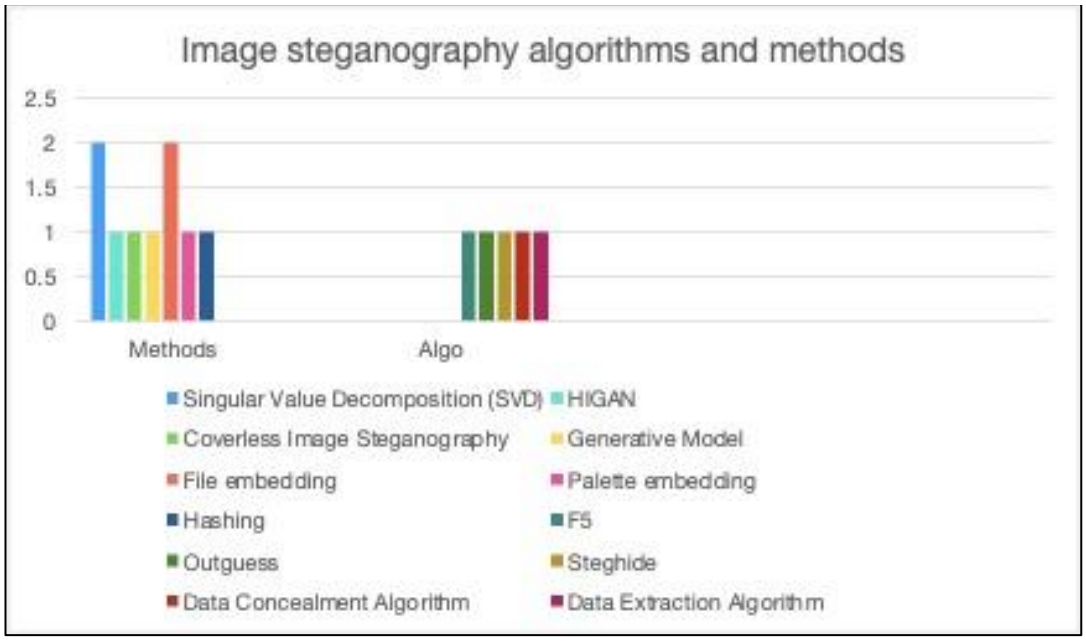
Note:

2. Image Steganography (Table 1) :

The DWT and DCT have wide applications in steganography (Graph 2). DWT decomposes the carrier signal into a number of frequency bands; it then becomes possible to embed data within factors that are less perceptible, hence, the visual or auditory effect is reduced. On the other hand, DCT applies focus of signal in little coefficients; effectively, it can be utilized to hide data in the frequency domain without affecting signal quality. Both techniques have high embedding capacity and robustness against steganalysis and are good for a multimedia steganography technique.

Although the simplest, LSB embedding remains one of the most popular methods in image steganography. It substitutes least significant bits of the pixel values with hidden data, an exchange that is almost imperceptible in many cases. While

it might not be as robust and capacious as DWT with DCT, the efficiency of LSB is certainly enough for most basic applications. It can be further combined with some encryption or error-correction techniques to provide better security.



Graph 3: Comparision and analysis of Image Steganography algorithms and methods.

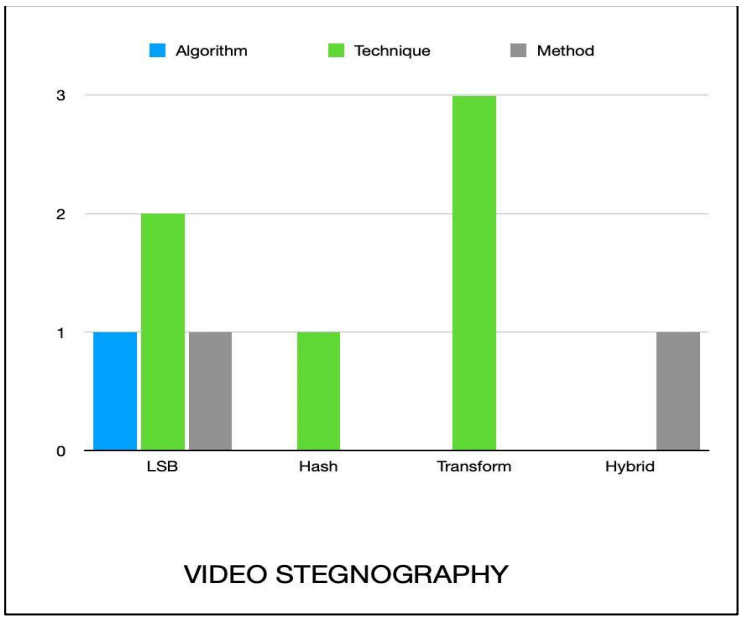
Note:

Compared to these, DFT is somewhat less widely applicable due to its high computational complexity and the risk of perceptual distortion. Such techniques are less well known and less used compared to their counterparts for JPEG and MP3 compression. All in all, while DWT and DCT account for most of the steganographic applications, LSB and DFT could come into application for some peculiarly oriented classes of them. SVD and file embedding are among the latest approaches to image steganography, which embed information within a digital image efficiently.

In SVD-based steganography, a matrix for an image is decomposed into singular values that are changed to embed the secret. The method, with its highly embedding capacity and good robustness against detection, has been quite inviting in applications of covert communications and data hiding (Graph 3).

File embedding, also referred to as payload embedding, embeds hidden data into the structure of the image. It typically involves appending or embedding files to the pixels of the cover image. Although not comparable in its capacity to SVD, it is, nonetheless, a quite simplistic approach for large volumes of data embedding.

Other image steganography methods are also of paramount importance, such as hybrid techniques that combine some of them. Basically, the choice depends on a number of factors, including the embedding capacity, robustness, security, and special requirements of the applications. In fact, understanding the strengths and weaknesses of each one makes steganographers pick techniques suitable for the objectives.



Graph 4: Comparision and analysis of Video Steganography algorithms, techniques and methods.

Note:

3. Video Steganography (Table 2):

Of all the transform techniques, DFT and DCT have been very popular for steganography, for they have shown to be effective means of concealing information without much perceptual distortion (Graph 4). Methods modulate a carrier in those frequency components less perceived by the human eye or ear, thus keeping the quality of the visual or acoustic signal; that is what makes them pretty fit for embedding of huge amounts of data within multimedia files like audio and video.

The other widely used techniques are LSB algorithms. They embed data in a cover medium by replacing its least significant bits of the pixel values, hence affecting the smallest visual alteration. Not being robust against sophisticated attacks that could be performed with steganalysis, which tries to detect changes in the images' statistical properties, the LSB algorithms are quite simple and fast. Nevertheless, owing to its convenience, the applicability of LSB has been extended to simple cases.

Recently, much attention has been paid to hybrid methods that incorporate several steganographic techniques. Such steganographic methods typically incorporate transform techniques with LSB embedding, encryption, or error correction coding to achieve improved capacity and robustness, and provide better security. Hybrid methods are in current use as they can achieve a good trade-off between capacity, security, and detection resistance for digital watermarking, content protection, and covert communications in very hostile environments.

XI. CONCLUSIONS

This paper, therefore, reviews in detail several techniques, algorithms, and methods of steganography used in embedding information in different media. Approaches are compared and contrasted to bring out the strengths and limitations that each of them possesses—a technique which involves the substitution of the Least Significant Bit, Discrete Cosine Transform, and Discrete Wavelet Transform. The evaluation, indeed, gives prominence to the fact that although LSB is still in use due to its simplicity, more sophisticated techniques like DCT and DWT ensure advanced security with strength against detection and attacks. The marriage between machine learning and artificial intelligence in steganography has been a trend of late; therefore, much more advanced and adaptive concealment methods are under way.

The steganographic technique should thus be chosen based on the imperceptibility, payload capacity, and resistance to attacks that an application may require. This survey is designed not only to provide detailed information about the different existing methods but also to bring awareness of the continuous research and innovation needed in the area. No doubt, future development will be oriented towards more efficient, safe, and versatile steganographic systems that will have applicability in a large number of situations, such as secure communications and digital watermarking.

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